

Numpy Problems

Scenario 1: Creating Feature & Target Vectors

You are setting up data for a simple regression model.

1. Create a 1D array `x` with values from 1 to 10.
2. Create a 1D array `y` of 10 elements containing all 1's (representing a bias term).
3. Create a 3 x 3 matrix of all zeros.

Scenario 2: Generating Synthetic Sensor Data

You need to simulate time values for an experiment.

1. Create an array `time_steps` of 5 evenly spaced values between 0 and 2.
2. Create an array `ticks` with numbers from 0 to 20 stepping by 5.

Scenario 3: Checking Array Dimensions

You loaded an array and need to check its structure before passing it to a function.

1. Create a 2D array `matrix` with shape (2, 3).
2. Print its shape, number of dimensions, and total number of elements using NumPy attributes.

Scenario 4: Reshaping a Flat List for Model Input

A model requires a 2D column vector, but you have a flat 1D array of 6 numbers: [10, 20, 30, 40, 50, 60].

1. Reshape this 1D array into a matrix with 3 rows and 2 columns.
2. Reshape the original 1D array into a column vector of shape (6, 1) using -1.

Scenario 5: Extracting Specific Features (Slicing)

Given a 3 x 3 matrix representing 3 samples and 3 features:

```
\begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{bmatrix}
```

1. Extract the first row.
2. Extract the last column.
3. Extract the top-left 2 x 2 sub-matrix.

Scenario 6: Filtering Outlier Values

You have an array of measured temperatures: $[22, 25, 19, 35, 21, 38, 24]$.

1. Create a boolean mask for temperatures strictly greater than 30 .
2. Filter out and display only the temperatures greater than 30 .

Scenario 7: Replacing Invalid Scores

You have an array of test scores: $[85, -1, 90, -1, 75]$, where -1 represents a missing test score. Replace all -1 values with 0 using `np.where` .

Scenario 8: Normalizing Inputs (Element-wise Operations)

You have a feature vector $x = [10, 20, 30, 40]$.

1. Multiply every element by 2 .
2. Square every element.
3. Subtract 5 from every element.

Scenario 9: Computing Summary Statistics

You have a matrix of student grades across 3 subjects (rows = students, columns = subjects):

$[[80, 90, 70], [60, 75, 85]]$

1. Compute the overall average grade across all students and subjects.
2. Compute the average grade for each subject (column-wise mean).

Scenario 10: Matrix Multiplication

Calculate the prediction of a simple model $y = X \cdot w$.

1. Given feature matrix $x = [[1, 2], [3, 4]]$ and weights $w = [0.5 \ 1.5]$.
2. Perform matrix multiplication between x and w .